

Ripple Carry Adder (RCA)

1. Introduction

A **Ripple Carry Adder (RCA)** is a basic digital circuit used to add two binary numbers. It consists of a sequence of full adders, each responsible for adding one bit of the inputs and a carry from the previous stage. The name *ripple carry* comes from the way the carry propagates through each adder stage — it "ripples" from the least significant bit (LSB) to the most significant bit (MSB).

2. Key Components

- **Full Adder Units:** Each bit is added using a full adder that takes two inputs and a carry-in.
- **Carry Propagation:** The carry-out from each full adder is passed as carry-in to the next higher-order full adder.

3. Structure

Let the two binary numbers be:

$$A = A_{n-1}A_{n-2} \dots A_1A_0, \quad B = B_{n-1}B_{n-2} \dots B_1B_0$$

Each full adder computes:

- **Sum:** $S_i = A_i \oplus B_i \oplus C_i$
- **Carry-out:** $C_{i+1} = (A_i \cdot B_i) + (B_i \cdot C_i) + (A_i \cdot C_i)$

The output carry from one stage becomes the input carry of the next stage. This continues from the LSB to the MSB.

4. Block Diagram (4-bit RCA)

The 4-bit Ripple Carry Adder includes:

- 4 Full Adder blocks (FA0 to FA3)
- Carry-in to FA0 is usually set to 0
- Carry-out of FA3 is the final carry

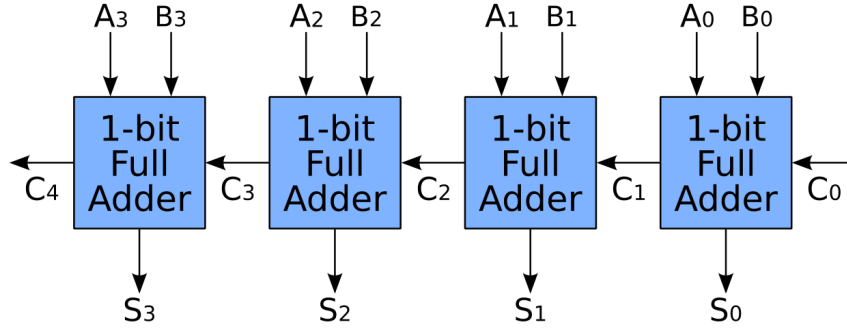


Figure 1: Block Diagram of a 4-bit Ripple Carry Adder

5. Full Adder Logic (Per Bit)

Each full adder uses the following logic:

$$\text{Sum (S)} = A \oplus B \oplus C_{in}$$

$$\text{Carry-out (Cout)} = (A \cdot B) + (B \cdot C_{in}) + (A \cdot C_{in})$$

This logic is repeated for each bit of the input.

6. Example

Let $A = 1101$ and $B = 1011$. The operation of the 4-bit RCA is as follows:

Bit Position	A	B	Cin	Sum	Cout
0 (LSB)	1	1	0	0	1
1	0	1	1	0	1
2	1	0	1	0	1
3 (MSB)	1	1	1	1	1

Result: 10000 (5 bits, including final carry)

7. Advantages

- **Simple Design:** Easy to understand and implement using full adders.
- **Modular:** Easily scalable by chaining more full adders.
- **Cost Effective:** Requires minimal additional logic beyond the full adders.

8. Disadvantages

- **Carry Propagation Delay:** Each carry must propagate through all previous adders.
- **Not Suitable for High-Speed Applications:** Worst-case delay is proportional to number of bits ($O(n)$).
- **Timing Bottleneck:** Limits the overall speed of addition.

9. Applications

- **Arithmetic Logic Units (ALUs):** Used in small or simple ALUs.
- **Digital Systems:** Suitable for low-speed and low-bit applications.
- **Educational Use:** Ideal for teaching and learning binary addition and carry propagation.

10. Conclusion

The Ripple Carry Adder provides a straightforward and modular approach to binary addition. While its design is simple and hardware-efficient, the carry ripple effect introduces significant delay in large-bit additions. For high-performance systems, faster alternatives such as carry-lookahead adders are used. However, the RCA remains important in understanding the basics of digital arithmetic and for use in systems where simplicity is prioritized over speed.